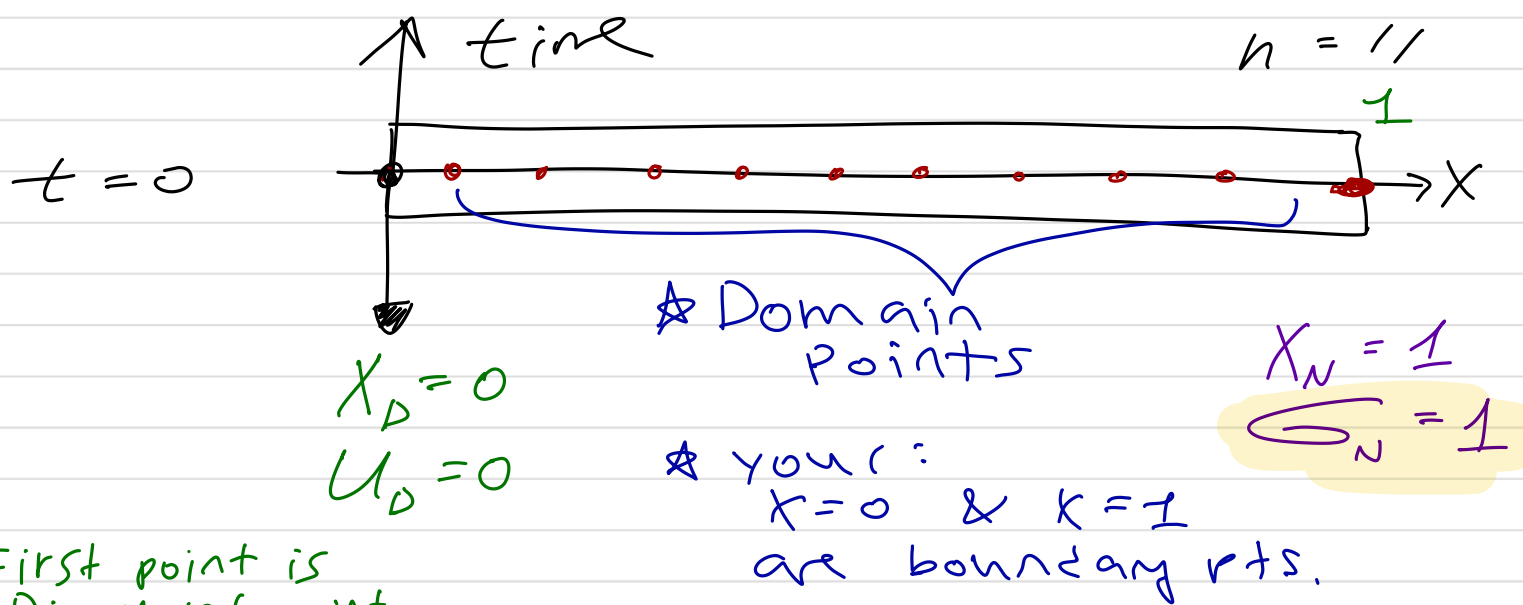




★ First point is Dirichlet point

★ last point $\rightarrow X = \text{linspace}(0, 1, n)$ is Neuman point

Input: $X_{\text{Domain}}, X_{\text{Neuman}}$

For 1D: $G = E \cdot \epsilon$

$\epsilon = \frac{\partial u}{\partial x}$

For 2D: $\vec{G} = [D] \cdot \vec{\epsilon}$

$\vec{\epsilon} = \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \end{bmatrix}$

$$(loss, gradients) = model(loss(net, X_{DIR}, X_{DOM}, X_{NEU}))$$

Pseudocode

Make predictions

$$\begin{aligned} U_{DOM} &= net(X_{DOM}) \\ U_N &= net(X_N) \\ U_{DIR} &= net(X_{DIR}) \end{aligned} \left. \vphantom{\begin{aligned} U_{DOM} &= net(X_{DOM}) \\ U_N &= net(X_N) \\ U_{DIR} &= net(X_{DIR}) \end{aligned}} \right\} \begin{array}{l} \text{★ predict separately like} \\ \text{shown, or use masks} \\ \text{to extract } U_N \text{ \& } U_{DIR} \end{array}$$

Note Elasticity:

★ When enforcing Strong PDE

we do not use boundary points.

★ Domain set is an open set!

Hooke's law

$$\underbrace{\nabla \cdot \underline{\underline{\sigma}}}_{\sigma_{ij,j}} + \underbrace{\rho \underline{\underline{b}}}_{\text{body loads}} = \underbrace{\rho \underline{\underline{a}}}_{\text{acceleration}} \text{ on } \Omega$$

$$\sigma_{xx,x} = 0$$

← Static 1D with no body loads

$$\sigma_{xx} = E \epsilon = E \frac{\partial u}{\partial x}$$

$$\sigma_{xx,x} = E \frac{\partial^2 u}{\partial x^2}$$

Enforce Domain

$$\frac{\partial U_{DOM}}{\partial x} = \text{gradient}(U_{DOM}) \leftarrow \text{use dl-gradient}$$

$$\frac{\partial^2 U_{DOM}}{\partial x^2} = \text{gradient}\left(\frac{\partial U_{DOM}}{\partial x}\right)$$

When doing Strong enforcement exclude bdry. pts from domain!

$$\mathcal{L}_D = \frac{1}{n_{DOM}} \sum \left(E \frac{\partial^2 U_{DOM}}{\partial x^2} \right)^2$$

★ since $\sigma_{xx,x} = 0$
 $\rightarrow E \cdot \frac{\partial^2 u}{\partial x^2} = 0$

we don't care for E here.

Enforce Boundary

Neumann

$$\begin{aligned} U_N &= net(X_N) \\ \frac{\partial U_N}{\partial x} &= \text{gradient}(U_N) \end{aligned}$$

Dirichlet

2 ways

try both

Stress at boundary

$$\sigma_N = E \cdot \frac{\partial u_N}{\partial x} \quad \star$$

$$L_N = \frac{1}{n_N} [\sigma_N - \bar{t} \star]$$

★ note how we did use E here but not for domain. we can choose to also not use it for neumann but that would mean setting $\bar{t} \star = \bar{t}/E$ where $\bar{t}=1$

strong & weak
Weak!

$$u_{\text{DIR}} = \text{het}(x_{\text{DIR}})$$

$$L_{\text{DIR}} = \frac{1}{n_{\text{DIR}}} (u_{\text{DIR}} - \bar{u}_{\text{DIR}})$$

here $\bar{u}_{\text{DIR}}$ is zero!

$$L_{\text{TOTAL}} = L_D + L_N + L_{\text{DIR}}$$

$$\frac{dL_{\text{TOTAL}}}{d \dots} = d \text{grad}(L_{\text{TOT}},$$

Net. Learnables)

W, b
weights biases

when training use
gradient descent
to update weights
& biases.

Strong Dirichlet

1st prediction.

$$U_{ALL} = \text{net}(X_{ALL})$$

$$U_{ALL}^* = U_{ALL} \cdot M_{\text{reverse}} + U_{DIR}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & \dots \end{pmatrix}$$

$$\vec{U}_{DIR} = \begin{bmatrix} \underbrace{\bar{U}_{DIR}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

★ also 0 in our case, but does not have to be.

$U_{ALL}^* \Rightarrow$ satisfies Dir. Hence Strong!
directly:

Side note: in FEM we do enforce our dirichlet boundary directly! ❗

$$0 = \underbrace{\int_{\check{V}} w \cdot (\text{PDE stuff}) dV}_{\check{V}} + \int_{\Gamma \rightarrow \Gamma_N} w \cdot (\text{BC stuff}) dA$$

We usually assume our solution u satisfies $u = \bar{u}$ directly on Γ_u , therefore we let $w = 0$ on $\Gamma_u \Rightarrow$ Galerkin! ❗